

SEMYACHIK, Russian Federation

WMO: 325090

Lat: 54.0822N

Lon: 159.9853E

Elev: 27

StdP: 101.00

Time Zone: 12.00 (E12)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-13.9	-12.3	-24.5	0.4	-12.2	-22.5	0.5	-11.0	23.9	-3.1	20.3	-3.5	5.0	320	0.783

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.4	20.2	14.6	18.2	14.2	16.6	13.4	16.1	18.5	15.0	17.1	14.1	15.9	4.5	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.1	10.7	17.1	14.1	10.1	15.8	13.2	9.5	14.8	45.1	18.5	42.2	17.1	39.7	15.9	21.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.9	15.1	12.8	DB	-16.2	25.0	3.0	3.1	-18.4	27.3	-20.2	29.1	-21.9	30.8	-24.1	33.1	(4)
(5)			WB	-17.1	18.2	2.7	1.7	-19.0	19.4	-20.6	20.4	-22.1	21.3	-24.0	22.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.8	-5.9	-5.8	-3.7	-0.4	3.8	8.2	12.1	13.4	10.7	5.6	-0.6	-4.6	(6)
(7)		DBStd	7.45	3.62	3.62	2.98	2.25	2.38	2.80	2.60	2.26	2.41	2.74	3.11	3.33	(7)
(8)		HDD10.0	2868	492	442	424	312	192	68	8	1	19	138	318	453	(8)
(9)		HDD18.3	5678	751	675	682	562	449	303	194	156	230	395	568	712	(9)
(10)		CDD10.0	234	0	0	0	0	1	15	73	105	40	1	0	0	(10)
(11)		CDD18.3	3	0	0	0	0	0	0	1	2	0	0	0	0	(11)
(12)		CDH23.3	18	0	0	0	0	0	2	9	6	0	0	0	0	(12)
(13)		CDH26.7	3	0	0	0	0	0	0	3	1	0	0	0	0	(13)
(14)	Wind	WSAvg	4.7	6.5	6.8	5.8	4.4	3.0	2.9	2.4	2.9	4.0	4.9	5.9	6.7	(14)
(15)	Precipitation	PrecAvg	1211	92	91	113	109	71	88	78	107	98	138	119	100	(15)
(16)		PrecMax	1737	214	370	269	360	200	262	202	207	221	328	355	334	(16)
(17)		PrecMin	786	10	16	22	17	20	5	5	18	27	21	23	13	(17)
(18)		PrecStd	219	51	76	54	77	40	62	48	57	46	85	73	78	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.7	1.6	2.8	8.3	15.2	21.6	23.8	23.8	20.0	13.8	6.7	3.6	(19)
(20)			MCWB	0.2	-0.3	0.0	3.2	8.2	13.2	16.2	15.5	13.3	7.9	3.9	2.7	(20)
(21)		2%	DB	0.9	0.3	1.4	5.2	11.2	16.7	19.7	20.4	17.5	11.6	5.6	2.5	(21)
(22)			MCWB	-0.1	-0.7	-0.8	1.5	6.5	11.7	15.3	14.9	12.3	6.8	4.0	1.5	(22)
(23)		5%	DB	0.3	-0.3	0.5	3.6	9.1	14.3	17.6	18.4	15.7	10.2	4.7	1.2	(23)
(24)			MCWB	-0.6	-1.2	-1.1	0.8	5.6	10.6	14.9	14.4	11.3	6.6	3.3	0.0	(24)
(25)		10%	DB	-0.4	-0.9	-0.2	2.6	7.5	12.2	15.9	16.7	14.2	9.2	3.8	0.3	(25)
(26)			MCWB	-1.3	-1.8	-1.3	0.2	4.9	9.8	14.0	13.9	10.8	6.2	2.2	-0.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.0	0.3	0.9	4.0	8.9	14.2	17.6	17.8	15.2	9.7	5.7	3.1	(27)
(28)			MCDB	1.2	0.6	1.8	7.1	13.4	19.2	21.4	20.9	17.6	10.9	5.9	3.3	(28)
(29)		2%	WB	0.2	-0.3	0.1	2.2	7.3	12.4	16.2	16.2	13.4	8.7	4.7	1.8	(29)
(30)			MCDB	0.6	0.1	0.5	4.1	10.2	15.8	18.7	18.5	15.8	10.0	5.0	2.4	(30)
(31)		5%	WB	-0.3	-0.8	-0.4	1.4	6.2	11.1	15.1	15.2	12.4	7.9	3.7	0.5	(31)
(32)			MCDB	0.2	-0.4	0.0	2.9	8.3	13.5	17.1	17.2	14.4	9.1	4.3	1.0	(32)
(33)		10%	WB	-1.1	-1.6	-0.9	0.7	5.2	10.0	14.0	14.4	11.6	7.0	2.7	-0.5	(33)
(34)			MCDB	-0.5	-1.0	-0.4	1.9	7.0	11.7	15.7	16.1	13.3	8.3	3.6	0.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	3.7	3.9	4.0	4.0	4.5	4.4	4.2	4.4	5.4	4.8	4.1	3.7	(35)
(36)			MCDBR	2.7	2.9	4.0	5.7	8.1	9.1	8.3	8.3	7.9	6.1	4.4	3.3	(36)
(37)			MCWBR	2.7	2.9	3.3	3.8	5.0	5.1	4.7	4.7	4.7	4.3	4.3	3.7	(37)
(38)		5% WB	MCDBR	2.5	2.7	2.9	4.8	7.0	7.6	6.9	6.5	6.5	4.8	3.7	3.3	(38)
(39)			MCWBR	2.8	3.0	2.9	3.8	4.9	4.8	4.4	4.5	4.8	4.4	4.4	4.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.292	0.322	0.374	0.423	0.479	0.466	0.475	0.446	0.367	0.348	0.305	0.283	(40)	
(41)		taud	2.130	2.110	1.973	1.904	1.855	2.076	2.099	2.178	2.396	2.287	2.184	2.111	(41)	
(42)		Ebn at Noon	636	745	777	791	767	787	770	769	800	715	614	557	(42)	
(43)		Edh at Noon	67	96	138	169	188	153	147	129	91	81	63	55	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.78	1.53	2.87	4.39	4.94	5.14	4.91	3.95	3.20	1.90	0.98	0.60	(44)	
(45)		RadStd	0.09	0.21	0.29	0.44	0.54	0.49	0.41	0.37	0.28	0.18	0.09	0.07	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page